

$$2.5 / 6, 16, 21, 31$$

$$6. \text{令 } Ux = y$$

$$\begin{cases} Ly = b \\ Ux = y \end{cases}$$

$$[L \ b] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ -3 & 1 & 0 & 0 & -2 \\ 3 & -2 & 1 & 0 & -1 \\ -5 & 4 & -1 & 1 & 2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} = [I \ y]$$

$$\therefore y = \begin{bmatrix} 1 \\ 1 \\ -2 \\ 1 \end{bmatrix} \quad [U \ y] = \begin{bmatrix} 1 & 3 & 4 & 0 & 1 \\ 0 & 3 & 5 & 2 & 1 \\ 0 & 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = [I \ x]$$

$$\therefore x = \begin{bmatrix} 3 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$16. \begin{bmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 14 & -10 \\ 0 & -14 & 10 \\ 0 & 21 & -15 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 2 & -6 & 6 \\ 0 & -7 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ \frac{3}{2} & -2 & 1 & 0 & 0 \\ -3 & 2 & 0 & 1 & 0 \\ 4 & 3 & 0 & 0 & 1 \end{bmatrix}$$

$$21. \quad A = BC$$

∵ B 可逆 ∴ B 可以行变换为 I

设这一系列行变换对应的矩阵为

$E_n, \dots, E_1, E_0$, 则

$$E_n \dots E_1 E_0 B = I$$

那么对 A 施加这些行变换:

$$E_n \dots E_1 E_0 A = E_n \dots E_1 E_0 BC = IC = C$$

∴ A 可以仅用行变换化为 C

$$-\frac{1}{4} \quad -\frac{1}{4}$$

$$31. \quad \left[\begin{array}{cccc|cccc} \textcircled{4} & -1 & -1 & & & & & \\ -1 & 4 & 0 & -1 & & & & \\ -1 & 0 & 4 & -1 & -1 & & & \\ & -1 & -1 & 4 & 0 & -1 & & \\ & & & & & & \dots & \\ & & & & & & & -1 & -1 & 4 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|cccc} 4 & -1 & -1 & & & & & \\ 0 & \textcircled{3.75} & -0.25 & -1 & & & & \\ 0 & -0.25 & 3.75 & -1 & -1 & & & \\ & -1 & -1 & 4 & 0 & -1 & & \\ & & & & & & & -1 & -1 & 4 \end{array} \right] \dots$$

$$U = \left[\begin{array}{cccc|cccc} 4 & -1 & -1 & & & & & \\ & 3.75 & -0.25 & -1 & & & & \\ & & 3.7333 & -1.0667 & -1 & & & \\ & & & 3.4286 & -0.2857 & -1 & & \\ & & & & 3.7083 & -1.0833 & -1 & \\ & & & & & 3.3919 & -0.2921 & -1 \\ & & & & & & 3.7052 & -1.0861 \\ & & & & & & & 3.3868 & -1 \end{array} \right]$$

$$L = \begin{bmatrix} 1 & & & & & & \\ -0.25 & 1 & & & & & \\ -0.25 & -0.066 & 1 & & & & \\ & -0.266 & -0.285 & 1 & & & \\ & & -0.267 & -0.083 & 1 & & \\ & & & -0.291 & -0.292 & 1 & \\ & & & & -0.269 & -0.086 & 1 \\ & & & & & -0.294 & -0.293 & 1 \end{bmatrix}$$

$$\text{令 } Ux = y$$

$$Ly = b$$

$$\text{解得 } x = \begin{bmatrix} 3.9569 \\ 6.5885 \\ 4.2392 \\ 7.3971 \\ 5.6029 \\ 8.7608 \\ 9.4115 \\ 12.0431 \end{bmatrix}$$

$$CA^{-1} = \begin{bmatrix} 0.2953 & 0.0866 & 0.0945 & 0.0509 & 0.0318 & 0.0227 & 0.0010 & 0.0082 \\ 0.0866 & 0.2953 & 0.0509 & 0.0945 & 0.0227 & 0.0318 & 0.0082 & 0.01 \\ 0.0945 & 0.0509 & 0.3271 & 0.1093 & 0.1045 & 0.0591 & 0.0318 & 0.0227 \\ 0.0509 & 0.0945 & 0.1093 & 0.3271 & 0.0591 & 0.1045 & 0.0227 & 0.0318 \\ 0.0318 & 0.0227 & 0.1045 & 0.0591 & 0.3271 & 0.1093 & 0.0945 & 0.0509 \\ 0.0227 & 0.0318 & 0.0591 & 0.1045 & 0.1093 & 0.3271 & 0.0509 & 0.0945 \\ 0.0010 & 0.0082 & 0.0318 & 0.0227 & 0.0945 & 0.0509 & 0.2953 & 0.0866 \\ 0.0082 & 0.01 & 0.0227 & 0.0318 & 0.0509 & 0.0945 & 0.0866 & 0.2953 \end{bmatrix}$$

3.1 13, 14

因教材上没有介绍“快速算法”这一算法，
在网上查找也并不知道“快速算法”指代哪
一种算法，目前理解为余因子展开算法。

13. 定义展开：

$$\begin{aligned}
 & \begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix} = 4 \begin{vmatrix} 0 & 2 & 0 & 0 \\ 3 & -6 & 4 & -8 \\ 0 & 5 & 2 & -3 \\ 0 & 9 & -1 & 2 \end{vmatrix} + (-7) \begin{vmatrix} 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots \\ 1 & 1 & 0 & 0 \end{vmatrix} + (-1) \cdot 3 \begin{vmatrix} 0 & 0 & 2 & 0 \\ 7 & 3 & -6 & -8 \\ 5 & 0 & 5 & -3 \\ 0 & 0 & 9 & 2 \end{vmatrix} \\
 & + (-5) \begin{vmatrix} 0 & 0 & 2 & 0 \\ 7 & 3 & -6 & 4 \\ 5 & 0 & 5 & 2 \\ 0 & 0 & 9 & -1 \end{vmatrix} \\
 & = 4 \left((-1) \cdot 2 \cdot \begin{vmatrix} 3 & 4 & -8 \\ 0 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix} \right) + (-1) \cdot 3 \left(2 \cdot \begin{vmatrix} 7 & 3 & -8 \\ 5 & 0 & -3 \\ 0 & 0 & 2 \end{vmatrix} \right) \\
 & + (-5) \cdot \left(2 \cdot \begin{vmatrix} 7 & 3 & 4 \\ 5 & 0 & 2 \\ 0 & 0 & -1 \end{vmatrix} \right) = -8 \cdot 3 + (-3) \cdot 2 \cdot (-30) + (-5) \cdot 30 \\
 & = 6
 \end{aligned}$$

余子式展开：

$$\begin{aligned}
 & \begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix} = 2 \cdot (-1) \cdot \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & 8 \\ 5 & 0 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{vmatrix} = 2 \cdot (-1) \cdot 3 \begin{vmatrix} 4 & 3 & -5 \\ 5 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix} \\
 & = -6 \cdot (-1) = 6
 \end{aligned}$$

14 定义展开:

$$\begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix}$$

$$= 6 \begin{vmatrix} 0 & -4 & 1 & 0 \\ -5 & 6 & 7 & 1 \\ 0 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 \end{vmatrix} + (-1) \cdot 3 \begin{vmatrix} 9 & -4 & 1 & 0 \\ 8 & 6 & 7 & 1 \\ 2 & 0 & 0 & 0 \\ 4 & 3 & 2 & 0 \end{vmatrix} +$$

$$2 \cdot \begin{vmatrix} 9 & 0 & 1 & 0 \\ 8 & -5 & 7 & 1 \\ 2 & 0 & 0 & 0 \\ 4 & 2 & 2 & 0 \end{vmatrix} + 4(-1) \cdot \begin{vmatrix} 9 & -4 & 0 \\ 8 & -5 & 6 & 1 \\ 2 & 0 & 0 & 0 \\ 4 & 2 & 3 & 0 \end{vmatrix}$$

$$= (-3) \cdot 2 \cdot \begin{vmatrix} -4 & 1 & 0 \\ 6 & 7 & 1 \\ 3 & 2 & 0 \end{vmatrix} + 2 \cdot 2 \cdot \begin{vmatrix} 0 & 1 & 0 \\ 5 & 7 & 1 \\ 2 & 2 & 0 \end{vmatrix} + (-4) \cdot 2 \cdot \begin{vmatrix} 0 & -4 & 0 \\ 5 & 6 & 1 \\ 2 & 3 & 0 \end{vmatrix}$$

$$= (-3) \cdot 2 \cdot (-1) \cdot (-11) + 4(-1)(-2) + (-8) \cdot 8$$

$$= 6$$

余因子展开:

$$\begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix}$$

$$= (-1) \cdot 2 \cdot \begin{vmatrix} 3 & 2 & 4 & 0 \\ 0 & -4 & 1 & 0 \\ -5 & 6 & 7 & 1 \\ 2 & 3 & 2 & 0 \end{vmatrix} = (-1) \cdot 2 \cdot (-1) \cdot 1 \cdot \begin{vmatrix} 3 & 2 & 4 \\ 0 & -4 & 1 \\ 2 & 3 & 2 \end{vmatrix}$$

$$= 2 \cdot 3 = 6$$

$$3.2 / 10. 13. 26. 27. 40$$

$$10. \begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -2 & -6 \\ -2 & -6 & 2 & 3 & 10 \\ 1 & 5 & -6 & 2 & -3 \\ 0 & 2 & -4 & 5 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & 0 & 3 & 6 \\ 0 & 2 & -5 & 2 & -1 \\ 0 & 2 & -4 & 5 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & 0 & 3 & 6 \\ 0 & 0 & -1 & 4 & 5 \\ 0 & 0 & 0 & 7 & 15 \end{vmatrix}$$

$$= -3 \begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & -1 & 4 & 5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 7 & 15 \end{vmatrix} = -3 \begin{vmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -2 & -6 \\ 0 & 0 & -1 & 4 & 5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{vmatrix} = -3 \cdot 1 \cdot 2 \cdot (-1) \cdot 1 \cdot 1 = 6$$

$$13. \begin{vmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{vmatrix} = (-1) \begin{vmatrix} 0 & -3 & -2 \\ 6 & -2 & -4 \\ -6 & 7 & 7 \end{vmatrix}$$

$$= (-1) \begin{vmatrix} 0 & -3 & -2 \\ 6 & -2 & -4 \\ 0 & 5 & 3 \end{vmatrix} = (-1) (-1) \cdot 6 \begin{vmatrix} -3 & -2 \\ 5 & 3 \end{vmatrix} = (-1) \cdot (-1) \cdot 6 \cdot 1 = 6$$

$$26. \begin{vmatrix} 3 & 2 & -2 & 0 \\ 5 & -6 & -1 & 0 \\ -6 & 0 & 3 & 0 \\ 4 & 7 & 0 & -2 \end{vmatrix} = -2 \begin{vmatrix} 3 & 2 & -2 \\ 5 & -6 & -1 \\ -6 & 0 & 3 \end{vmatrix} = -2 \cdot 0 = 0$$

11行列为0, 矩阵不可逆, 11组向量线性有关。

27. a. $\checkmark$

b. $\times$ 还有可能经过行缩放, 这同样会改变行列式的值.

c. $\checkmark$

d. $\times$ $\det(A+B) \neq \det A + \det B$

40 a. $\det AB = \det A \cdot \det B = 3$

b. $\det B^5 = (\det B)^5 = -1$

c. $\det 2A = 2^4 \det A = -48$

$$\det A^T B A = \det A^T \cdot \det B \cdot \det A = -9$$

$$\det B^{-1} A B = \frac{1}{\det B} \cdot \det A \cdot \det B = -3$$